

lqb program no 6.

Write a C program to simulate the following contiguous memory allocation techniques. a) Worst-fit

b) Best-fit

c) First-fit

```
#include <stdio.h>
```

```
void firstFit(int blockSize[], int m, int processSize[], int n)
```

```
{
    int allocation[10];

    for(int i = 0; i < n; i++)
        allocation[i] = -1;

    for(int i = 0; i < n; i++)
    {
        for(int j = 0; j < m; j++)
        {
            if(blockSize[j] >= processSize[i] && blockSize[j] != -1)
            {
                allocation[i] = j;

                blockSize[j] = -1;

                break;
            }
        }
    }

    printf("\n--- First Fit ---\n");
    printf("Process No.\tProcess Size\tBlock No.\n");

    for(int i = 0; i < n; i++)
    {
        printf("%d\t\t%d\t\t", i + 1, processSize[i]);

        if(allocation[i] != -1)
            printf("%d\n", allocation[i] + 1);
        else
            printf("Not Allocated\n");
    }
}
```

```

void bestFit(int blockSize[], int m, int processSize[], int n)
{
    int allocation[10];

    for(int i = 0; i < n; i++)
        allocation[i] = -1;

    for(int i = 0; i < n; i++)
    {
        int best = -1;

        for(int j = 0; j < m; j++)
        {
            if(blockSize[j] >= processSize[i] && blockSize[j] != -1)
            {
                if(best == -1 || blockSize[j] < blockSize[best])
                    best = j;
            }
        }

        if(best != -1)
        {
            allocation[i] = best;

            // mark block as used
            blockSize[best] = -1;
        }
    }

    printf("\n--- Best Fit ---\n");
    printf("Process No.\tProcess Size\tBlock No.\n");

    for(int i = 0; i < n; i++)
    {
        printf("%d\t%d\t", i + 1, processSize[i]);

        if(allocation[i] != -1)
            printf("%d\n", allocation[i] + 1);
        else
            printf("Not Allocated\n");
    }
}

```

```

void worstFit(int blockSize[], int m, int processSize[], int n)
{
    int allocation[10];

    for(int i = 0; i < n; i++)
        allocation[i] = -1;

    for(int i = 0; i < n; i++)
    {
        int worst = -1;

        for(int j = 0; j < m; j++)
        {
            if(blockSize[j] >= processSize[i] && blockSize[j] != -1)
            {
                if(worst == -1 || blockSize[j] > blockSize[worst])
                    worst = j;
            }
        }

        if(worst != -1)
        {
            allocation[i] = worst;

            // mark block as used
            blockSize[worst] = -1;
        }
    }

    printf("\n--- Worst Fit ---\n");
    printf("Process No.\tProcess Size\tBlock No.\n");

    for(int i = 0; i < n; i++)
    {
        printf("%d\t%d\t", i + 1, processSize[i]);

        if(allocation[i] != -1)
            printf("%d\n", allocation[i] + 1);
        else
            printf("Not Allocated\n");
    }
}

int main()

```

```

{
    int m, n;

    int blockSize[10], processSize[10];
    int first[10], best[10], worst[10];

    printf("Enter number of memory blocks: ");
    scanf("%d", &m);

    printf("Enter sizes of %d memory blocks:\n", m);

    for(int i = 0; i < m; i++)
    {
        scanf("%d", &blockSize[i]);

        first[i] = blockSize[i];
        best[i] = blockSize[i];
        worst[i] = blockSize[i];
    }

    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter sizes of %d processes:\n", n);

    for(int i = 0; i < n; i++)
    {
        scanf("%d", &processSize[i]);
    }

    firstFit(first, m, processSize, n);
    bestFit(best, m, processSize, n);
    worstFit(worst, m, processSize, n);

    return 0;
}

```

```
"C:\Users\sahne\OneDrive\De x + v
Enter sizes of memory blocks:
100 500 200 300 600
Enter number of processes: 4
Enter sizes of processes:
212 417 112 426

FIRST FIT Allocation:
Process No    Process Size    Block No
1             212             2
2             417             5
3             112             2
4             426             Not Allocated

BEST FIT Allocation:
Process No    Process Size    Block No
1             212             4
2             417             2
3             112             3
4             426             5

WORST FIT Allocation:
Process No    Process Size    Block No
1             212             5
2             417             2
3             112             5
4             426             Not Allocated

Process returned 0 (0x0)  execution time : 51.396 s
Press any key to continue.
```